Hexaphotone: The Six-Dimensional History of Light

Manuel Julin

Version 3.1 - Revised Edition
May 25, 2026

Abstract

This document presents a revised, self-contained formulation of the Hexaphotone framework: an information-centric classical wave-optics model representing the informational state of an electromagnetic radiation element by a six-component vector $\mathbf{H} = (E, f, P_i, \phi, \vec{p}, \vec{d})$. The framework is explicitly classical; quantum extensions are outlined as future work. Light fields are modelled as superpositions of information-carrying modes; interactions with matter are represented as linear operators on modal coefficient spaces. A canonical Gaussian mapping from the Hexaphoton vector to modal amplitudes is defined and connected to classical scattering theory (Mie and T-matrix). Explicit physical constraints between the six components are introduced. Numerical pipeline, validation metrics, and experimentally testable protocols including one prediction specific to the Hexaphotone framework are specified to ensure reproducibility.

Contents

1 Introduction

Classical electrodynamics describes light fields through Maxwell's equations and their solutions in various bases (plane waves, multipoles, Gaussian beams). This description can obscure the informational content of a field: which degrees of freedom carry the source signature, and how does matter transform them into observable patterns?

The Hexaphotone framework reframes six standard electromagnetic field descriptors as a compact information vector, emphasizing modal structure and linear operator action. The framework is modular:

- A source representation: the Hexaphoton ensemble $\{H_i\}$.
- A modal mapping F: produces incident modal amplitudes $a_k(H)$.
- A material/operator model T^M: transforms incident into outgoing modes.

Scope (v3.1): The present formulation is a classical wave-optics framework. It does not quantize the field and makes no claims about single-photon quantum effects. Quantum extensions are described in Section 8 as future work.

2 Formal Definitions and Component Constraints

2.1 The Hexaphoton Vector

The Hexaphoton vector is defined as:

$$\mathbf{H} = (E, f, P_i, \phi, \vec{p}, \vec{d})$$

where the components are described in Table 1.

Table 1: Components of the Hexaphoton vector **H**.

Component	Symbol	Description
Energy scale	E	Proportional to total modal energy; $E > 0$.
Frequency	f	Central frequency or spectral centroid; $f > 0$.
Polarization	P_i	Jones vector (coherent) or Stokes parameters (partial).
Phase	ϕ	Global or reference phase; $\phi \in [0, 2\pi)$.
Momentum	$\vec{p}$	Modal momentum vector (see constraints below).
Direction	$\vec{d}$	Unit propagation direction; $\lvert\vec{d}\rvert = 1$.

2.2 Physical Constraints Between Components

The six components are not fully independent. In the classical electromagnetic limit, the following constraints must hold:

- **Energy-frequency constraint:** For a quasi-monochromatic beam:

  $$E = h_{\text{eff}} \cdot f \cdot N_{\text{eff}}$$

  where h_{eff} is an effective action scale (equal to Planck's constant h for a photon-number interpretation) and N_{eff} is the effective mode occupation. E and f are coupled.

- **Momentum-frequency-direction constraint:**

  $$|\vec{p}| = \frac{h_{\text{eff}} \cdot f}{c} = \frac{2\pi h_{\text{eff}}}{\lambda}$$

  The direction of $\vec{p}$ must be consistent with $\vec{d}$: $\vec{p} = |\vec{p}| \cdot \vec{d}$. Thus, $\vec{p}$ is not an independent parameter; it is determined by f and $\vec{d}$. It is retained in **H** as a convenience variable; all implementations must enforce this constraint.

Design note: Retaining $\vec{p}$ explicitly is useful in scattering problems where multiple momentum components are superposed. The constraint applies to each monochromatic component; broadband ensembles integrate over f.

2.3 Field Representation as Modal Superposition

The electric field is represented as a superposition of modes:

$$\mathbf{E}(\mathbf{r}, t) = \sum_{k=0}^{\infty} a_k(t) u_k(\mathbf{r})$$

where $u_k(\mathbf{r})$ is an orthonormal set of spatial (and vector) modes, and $a_k(t)$ are complex modal amplitudes encoding amplitude, phase, and polarization.

The spatial Fourier transform of the field is:

$$\tilde{\mathbf{E}}(\mathbf{k}, t) = \mathcal{F}\{\mathbf{E}(\mathbf{r}, t)\}(\mathbf{k})$$

3 Operator Formalism for Interactions

3.1 Material Operator (General Linear Model)

The output field is related to the input field by a material operator:

$$\mathbf{E}_{\text{out}}(\mathbf{r}, t) = T_M[\mathbf{E}_{\text{in}}(\mathbf{r}, t)]$$

In the spatial-frequency representation:

$$\tilde{\mathbf{E}}_{\text{out}}(\mathbf{k}, t) = H_M(\mathbf{k}; f) \cdot \tilde{\mathbf{E}}_{\text{in}}(\mathbf{k}, t)$$

where $H_M(\mathbf{k}; f)$ is a complex, matrix-valued transfer function encoding geometry, material parameters, and frequency dependence.

3.2 Measurement as Projection

Measurement outcomes are inner products of the outgoing field with detector modes m_j:

$$I_j = \langle m_j, \mathbf{E}_{\text{out}} \rangle = \int m_j^*(\mathbf{r}) \cdot \mathbf{E}_{\text{out}}(\mathbf{r})\, d^3r$$

Apparent state reduction is interpreted as selection of modal channels by the measurement operator.

3.3 Operator Decomposition

The material operator can be decomposed as:

$$T_M = T_{\text{prop}} \circ T_{\text{surf}} \circ T_{\text{bulk}}$$

where the components represent free propagation, surface interactions (reflection/refraction), and bulk scattering. Each admits a representation in the chosen modal basis.

4 Mapping Hexaphoton to Modal Amplitudes (Canonical Form)

A mapping F assigns modal amplitudes to a Hexaphoton vector. In v3.1, a canonical Gaussian parameterization is defined as the reference implementation.

4.1 Canonical Gaussian Prototype in k-Space

The modal amplitude distribution is given by:

$$a(\mathbf{k}; \mathbf{H}) = A(E, f) \cdot P(P_i) \cdot \exp\left(\frac{-|\mathbf{k} - \mathbf{k}_0|^2}{2\sigma(f)^2}\right) \cdot e^{i\phi}$$

where:

- $\mathbf{k}_0 = \frac{2\pi f}{c}\vec{d}$ (determined by $\vec{d}$ and the constraint $|\mathbf{k}_0| = 2\pi f/c$).
- $\sigma(f) = \alpha \cdot f/c$: frequency-dependent spectral spread; α is calibrated to the source bandwidth.
- $A(E, f)$: normalization amplitude enforcing the energy constraint $\sum_k |a_k|^2 = C \cdot E$.
- $P(P_i)$: Jones-vector polarization weight for each vector mode component.
- ϕ: global phase factor via $e^{i\phi}$.

4.2 Projection onto Discrete Modal Basis

The modal amplitudes are projected onto a discrete basis:

$$a_k(\mathbf{H}) = \int u_k^*(\mathbf{k})a(\mathbf{k};\mathbf{H})\,d^2k$$

The mapping F_k is fully specified by $\{\alpha, C\}$ and the choice of modal basis. Alternative parameterizations define model variants; all must satisfy the constraints in Section **??**.

Falsifiability note: With the canonical form defined, the model has two free scalar parameters (α, C) fixed by at most two independent measurements before any further prediction is made. All further predictions are then parameter-free given that calibration.

5 Connection to Scattering Theory (Mie and T-Matrix)

5.1 Mie Theory as Sphere Reference Operator

For a homogeneous dielectric sphere (radius R, refractive index m), Mie theory yields multipole coefficients $a_n(f), b_n(f)$ and angular scattering amplitude $S(\theta, \phi; f)$. Under paraxial mapping, angular scattering maps to transverse wavevectors $\mathbf{k}_\perp$:

$$H_{\text{mie}}(\mathbf{k}; f) \approx \sum_{n=0}^{N} a_n(f)Y_n(\hat{k}), \quad k_\perp = \frac{2\pi}{\lambda}\sin\theta \quad \text{(paraxial)}$$

5.2 T-Matrix Generalization

The T-matrix maps incident multipole coefficients to scattered multipole coefficients for arbitrary scatterers, yielding a matrix representation of T^M in the modal basis:

$$a_{\text{out}}(\mathbf{k}) = H_M(\mathbf{k}) \cdot a_{\text{in}}(\mathbf{k})$$

with a_{in} produced by the Hexaphoton mapping F and H_M provided by Mie/T-matrix or a numerical solver.

6 Numerical Implementation and Reproducibility

6.1 Computational Pipeline

1. Define Hexaphoton ensemble $\{H_i\}$ and enforce constraints from Section **??**.
2. Construct canonical modal distributions $a(\mathbf{k}; H_i)$ using the Gaussian prototype (Section **??**).
3. Discretize $\mathbf{k}$-space; ensure Nyquist sampling for the spatial domain of interest.
4. Compute incident field via inverse Fourier transform of $a(\mathbf{k})$.
5. Compute object operator: for a sphere, use a Mie library to obtain $S(\theta, \phi)$; map to $H_M(\mathbf{k})$. For general objects, use T-matrix or numerical solvers (BEM, FEM, FDTD).
6. Apply operator: $a_{\text{out}}(\mathbf{k}) = H_M(\mathbf{k}) \cdot a_{\text{in}}(\mathbf{k})$.
7. Reconstruct outgoing field and compute observables $I(\mathbf{r}) = |\mathbf{E}(\mathbf{r})|^2$.

6.2 Validation Metrics

- Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(I_{\text{ref}}(\mathbf{r}_i) - I_{\text{model}}(\mathbf{r}_i)\right)^2$$

- Spectral overlap: normalized inner product of Fourier spectra.
- Polarization fidelity: angular difference and degree-of-polarization comparison.

6.3 Software Recommendations

- Numerical core: NumPy, SciPy, Matplotlib.
- Mie: `miepython` or equivalent validated library.
- T-matrix: available packages for spheroids and aggregates.
- Reproducibility: Jupyter notebooks, parameter YAML files, `requirements.txt` with pinned versions.

7 Experimental Predictions and Test Protocols

v3.1 update: Predictions A–C are standard wave-optics consistency checks. Prediction D is specific to the Hexaphotone framework.

7.1 Prediction A: Reflection Imaging of Shaped Surfaces

A shaped reflective surface illuminated by a Hexaphoton ensemble will produce a projected intensity distribution preserving contour information when T^M transmits the relevant modal channels.

- **Measurement:** Compare projected intensity with model predictions.
- **Role:** Consistency check with classical optics.

7.2 Prediction B: Slit Filtering and Source Dominance

A narrow aperture suppresses high spatial frequencies; transmitted patterns emphasize low-frequency components reflecting source structure.

- **Measurement:** Record transmitted intensity for different aperture geometries and compare Fourier spectra to incident modal distribution.
- **Role:** Classical diffraction consistency check.

7.3 Prediction C: Sphere Reference Test (Mie)

A homogeneous dielectric sphere acts as a known operator H_M^{mie}. For incident Hexaphoton distributions with controlled $\vec{d}$ and $\vec{p}$, the scattered angular distribution should match Mie-based predictions when the canonical mapping F is applied with calibrated (α, C).

- **Measurement:** Angularly resolved scattering and near-field intensity mapping.
- **Role:** Consistency check and parameter calibration.

7.4 Prediction D: Phase-Encoded Directional Discrimination (Framework-Specific)

Two Hexaphoton ensembles identical in $(E, f, P_i, \vec{p})$ but differing only in ϕ and $\vec{d}$ should produce distinguishable far-field patterns through an asymmetric aperture, because the canonical mapping encodes ϕ as a global phase shift altering interference with aperture boundary modes. Classical Fourier optics predicts the same intensity for global phase shifts. The Hexaphotone prediction differs if ϕ couples to modal weights in F—a falsifiable distinction.

- **Measurement:** Single-mode interferometric detection after asymmetric aperture; record visibility vs. $\Delta\phi$.
- **Role:** Framework-specific falsifiable test.

7.5 Protocol Essentials

- Use narrowband and broadband sources to separate spectral effects.
- Record polarization-resolved intensity.
- Calibrate detectors; correct for system transfer functions.
- Fix (α, C) using Predictions A–C before testing Prediction D.
- Provide raw data and processing scripts for reproducibility.

8 Discussion, Limitations and Extensions

The Hexaphotone framework is an information-centric modelling layer interfacing source characterization and classical scattering/propagation operators. With the canonical Gaussian mapping defined in v3.1, the framework is now falsifiable with two free parameters fixed by calibration.

8.1 Limitations

- Classical only: single-photon quantum effects require field quantization; outside present scope.
- Linear operators only: nonlinear and inelastic interactions are not covered.
- Constraint enforcement: implementations must explicitly enforce the E–f–$\vec{p}$ constraint (Section **??**).
- Modal basis dependence: numerical efficiency and interpretability depend on the choice of basis $\{u_k\}$.

8.2 Extensions

- Include orbital angular momentum (OAM) and topological mode indices as a 7th information channel.
- Extend F to stochastic ensembles to model partial coherence via the mutual coherence function.
- Promote modal amplitudes to annihilation/creation operators; define Hexaphoton states in Fock space.
- Machine-learning parameterization of F trained on experimental scattering data.

9 Conclusion

Hexaphotone Theory v3.1 formalizes an information-centric classical wave-optics representation of light and provides a modular operator framework relating source information to observable fields via established scattering theory. Key improvements over v3.0 include:

- Explicit physical constraints between the six Hexaphoton components.
- A canonical Gaussian modal mapping with two calibratable parameters.
- A framework-specific experimental prediction (Prediction D) falsifiable independently of classical Fourier optics.

The framework is modular and extensible for further theoretical and experimental development.

References

References

[1] Goodman, J.W. *Introduction to Fourier Optics*, 3rd ed. Roberts and Company, 2005. ISBN 978-0974707723.

[2] Born, M. and Wolf, E. *Principles of Optics*, 7th ed. Cambridge University Press, 1999. DOI: 10.1017/CBO9781139644181.

[3] Bohren, C.F. and Huffman, D.R. *Absorption and Scattering of Light by Small Particles.* Wiley-VCH, 1998. ISBN 978-0471293408.

[4] Mishchenko, M.I., Travis, L.D., Lacis, A.A. *Scattering, Absorption, and Emission of Light by Small Particles.* Cambridge University Press, 2002.

[5] Waterman, P.C. Matrix formulation of electromagnetic scattering. *Proceedings of the IEEE*, 53(8):805–812, 1965. DOI: 10.1109/PROC.1965.4058.

[6] Saleh, B.E.A. and Teich, M.C. *Fundamentals of Photonics*, 3rd ed. Wiley, 2019. ISBN 978-1119506874.

License

Ethical Use Statement

This project is licensed under CC-BY-SA 4.0. The author objects to any use of this work for the design, development, production or deployment of weapons or other instruments of harm.

A Central Equations

A.1 Hexaphoton Vector

$$\mathbf{H} = (E, f, P_i, \phi, \vec{p}, \vec{d})$$

A.2 Energy-Frequency Constraint

$$E = h_{\text{eff}} \cdot f \cdot N_{\text{eff}}$$

A.3 Momentum Constraint

$$|\vec{p}| = \frac{h_{\text{eff}} \cdot f}{c} = \frac{2\pi h_{\text{eff}}}{\lambda}, \quad \vec{p} = |\vec{p}| \cdot \vec{d}$$

A.4 Modal Expansion

$$\mathbf{E}(\mathbf{r}, t) = \sum_{k=0}^{\infty} a_k(t) u_k(\mathbf{r})$$

A.5 Fourier Transform

$$\tilde{\mathbf{E}}(\mathbf{k}, t) = \mathcal{F}\{\mathbf{E}(\mathbf{r}, t)\}(\mathbf{k})$$

A.6 Transfer Function

$$\tilde{\mathbf{E}}_{\text{out}} = H_M(\mathbf{k}; f) \cdot \tilde{\mathbf{E}}_{\text{in}}$$

A.7 Measurement Projection

$$I_j = \int m_j^*(\mathbf{r}) \cdot \mathbf{E}_{\text{out}}(\mathbf{r})\, d^3 r$$

A.8 Canonical Gaussian Mapping

$$a(\mathbf{k}; \mathbf{H}) = A(E, f) \cdot P(P_i) \cdot \exp\left(\frac{-|\mathbf{k} - \mathbf{k}_0|^2}{2\sigma(f)^2}\right) \cdot e^{i\phi}$$

A.9 Mode Projection

$$a_k(\mathbf{H}) = \int u_k^*(\mathbf{k}) a(\mathbf{k}; \mathbf{H})\, d^2 k$$

A.10 Operator Composition

$$T_M = T_{\text{prop}} \circ T_{\text{surf}} \circ T_{\text{bulk}}$$

A.11 Operator Application

$$a_{\text{out}}(\mathbf{k}) = H_M(\mathbf{k}) \cdot a_{\text{in}}(\mathbf{k})$$

A.12 Intensity

$$I(\mathbf{r}) = |\mathbf{E}(\mathbf{r})|^2$$

A.13 Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(I_{\text{ref}}(\mathbf{r}_i) - I_{\text{model}}(\mathbf{r}_i)\right)^2$$